

PID and LQR Solution Notes

Corrected implementation | Illustrative gains and explicit limitations

Use the corrected model handout and solutions/simulate_quarter_car.m. Parameters and tuning settings are shared through solutions/config.json. This document explains the controller construction; the supplied gains are not asserted to meet every benchmark.

1. Passive baseline

Set $u = 0$ and compare the resulting body motion, tire deflection, suspension travel and zero actuator effort against both active designs. The four-state mechanical plant is stable. Its slow decay can exceed the 10 s observation interval.

2. Ideal PID on suspension deflection

$$u = -K_p s - K_d s_{\dot{}} - K_i x_i, \quad x_{i\dot{}} = s$$

```
Kp = 208000; Kd = 832000; Ki = 624000
K = [Kp*C + Kd*C*A, Ki]
F = -Kd*C*Bw
u = -K*xa + F*zr
Acl = Aa - Bua*K; Bcl = Bwa + Bua*F
```

Since $C \cdot B_u = 0$, recovering $s_{\dot{}}$ introduces no algebraic control loop. The explicit road term is a coordinate-conversion term; a physical controller can use measured relative velocity instead of measuring road height.

Units: K_p in N/m, K_d in N s/m, and K_i in N/(m s). The example retains the original course gains. A 0.2 m ideal road step causes $s_{\dot{}}$ to jump to -9.3875 m/s and the derivative force to jump to about 7.81 MN. This illustrates an impractical ideal-PID response, not a feasible actuator design.

For a practical extension, use derivative filtering, smooth road transitions, actuator limits and anti-windup, then re-tune and re-evaluate the model. These extensions are not silently included in the linear reference.

3. Tuning workflow

Study $G_u = S/U$ with zero road disturbance. Use pidTuner or tune gains directly; verify the closed-loop road response separately. Always check closed-loop poles and all metrics. Minimizing suspension deflection can increase body acceleration and actuator force.

LQR and numerical evaluation

Integral state feedback and exact continuous-input simulation

4. LQR with integral action

```
J = integral (xa_transpose*Q*xa + u*R*u) dt
Q = diag([3e7, 8e7, 8e7, 2e7, 2e10]); R = 0.01
K = lqr(Aa, Bua, Q, R); u = -K*xa
```

The fourth Q entry penalizes η , not s_{dot} . Coordinate changes alter the meaning of the weights. The fifth weight penalizes accumulated suspension-deflection error. Increasing R discourages force; changing Q reallocates state penalties. The example assumes full-state availability and no saturation.

5. Simulation and energy integration

The linear closed-loop model is propagated exactly between output times using `expm`. For a step, append a constant road state and change it at $t = 1$ s. For a sine, append the oscillator states z_r and z_{r_dot} . This avoids replacing a continuous sine with a staircase input.

```
z_dot = H*z,      z(t+dt) = E*z(t),      E = expm(H*dt)
y = h*z
energy_interval = z(t)_transpose * G * z(t)
G = integral_0^dt expm(H_transpose*tau)*h_transpose*h
    *expm(H*tau) dtau
```

G is computed with a block matrix exponential. Sum interval energies and divide by T before taking a square root. This accounts for fast PID transients that would be misestimated by applying a trapezoid directly to a coarse sampled spike. Peak and recovery metrics remain sampled at 1 ms.

6. Interpreting the results

Compare the checked-in tables and figures for both road inputs. LQR reduces body acceleration relative to the passive baseline for these settings, but travel and force targets may still be missed. The ideal PID strongly suppresses suspension deflection while demanding excessive force. These conclusions concern the tested nominal parameters, not proven robustness.

Report suspension recovery in the stated ± 4 mm band. Body overshoot and body settling are separate supplemental metrics. Percentage overshoot relative to a zero final suspension deflection is undefined.

Reference results

Generated from the corrected model and shared configuration

Step input

Metric	Passive	PID	LQR
Body acceleration RMS (m/s ²)	2.498	13.28	0.7532
Tire deflection RMS (m)	0.01364	0.01845	0.00613
Peak suspension deflection (m)	0.2207	0.006251	0.196
Force RMS (kN)	0	33.17	3.05
Peak force (kN)	0	7810	33.31
Body overshoot (%)	95.35	54.31	3.205
Body settling time (s)	not settled	1.27	1.621
Suspension recovery time (s)	not settled	0.142	1.618

Sine input

Metric	Passive	PID	LQR
Body acceleration RMS (m/s ²)	1.593	0.8597	0.4499
Tire deflection RMS (m)	0.008302	0.004743	0.002419
Peak suspension deflection (m)	0.1176	0.001286	0.1044
Force RMS (kN)	0	2.141	5.384
Peak force (kN)	0	8.15	7.88
Body overshoot (%)	N/A	N/A	N/A
Body settling time (s)	N/A	N/A	N/A
Suspension recovery time (s)	N/A	N/A	N/A

RMS values span 10 s. "Not settled" means not within the specified band at the end of the record. Sampling affects reported peaks and recovery times; RMS values use exact interval integration. See solutions/results.json for full precision and VALIDATION.md for checks and runtime limitations.